KUMMER-N13: Concrete 16-bit State Replay

Source-bound research packet

25 October 2026

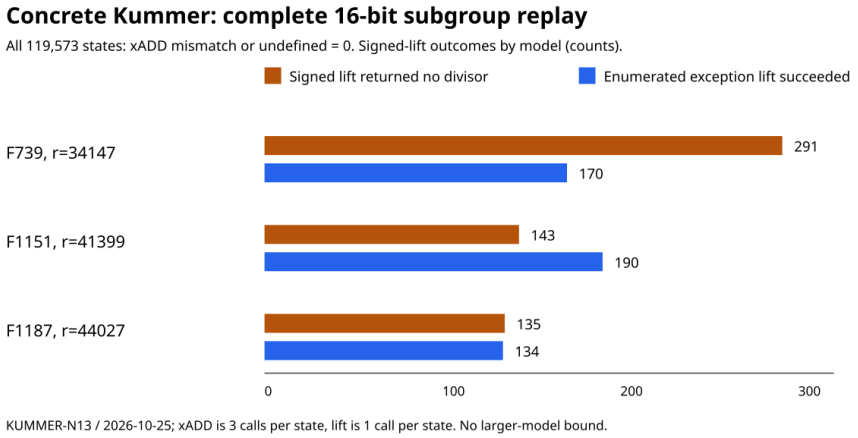
The actual cached theta xADD and signed-recovery source was replayed on every signed state in three frozen 16-bit odd-prime subgroups. The source produced three xADD outputs and one signed lift per state. An independent Jacobian oracle checked the outputs. This is finite exhaustion of these three groups, not a bound for the other 18 frozen models.

p	states	xADD calls	xADD errors	lift correct	lift no divisor
739	34,147	102,441	0	33,856	291
1,151	41,399	124,197	0	41,256	143
1,187	44,027	132,081	0	43,892	135
Total	119,573	358,719	0	119,004	569

Among correct lifts, the source's enumerated exceptional branch returned 170, 190, and 134 results by model, respectively. The 569 failures split into 108 `Delta_zero`, 108 `b1_coefficient_zero`, and 353 `general_conversion_inapplicable`. These are current implementation branch failures. They do not contradict the abstract pair-sign uniqueness theorem.

The lift calls used 41,887,801 M, 33,639,052 A, and 2,268,451 I in signed recovery after cached inverse setup. Exceptional enumeration added 92,925 field M, 1,482 field S, 3,677 field I, 169,982 field additions, and 741 Jacobian additions. These counters exclude setup, labels, collector work, relation processing, and end-to-end checks. No ratio to rho is measured.

Evidence: `results/states.csv` (one row per scalar), `results/summary.json` (source and CSV digests), and `SHA256SUMS`. Native `validate` recomputes counts and digests. The 18 larger models, all-state fallback proof, full target cohort, and matched rho pricing remain open.



Counts from the 119,573 per-state records; KUMMER-N13, 2026-10-25.